Water Usage Report

Thabo's Farm (demo) · all 2 zones · Sun 4 Oct to Sat 10 Oct
Generated 10 Oct 2026, 07:00 SAST

This week, do this

1 **Water Zone 1 today (Sat 10 Oct)**: about 66.4 minutes (about 996 L).
High confidence
Why: By the end of today your soil will be drier than tomato can stand, and no rain is expected today. (note 1)

2 **Water Zone 2 on Monday (Mon 12 Oct)**: about 51.9 minutes (about 779 L).
High confidence
Why: At the rate tomato is using water, the soil will reach its watering point in about 2 days. (note 2)

3 **Your tomato in Zone 1 got about 65% of what it needed this week**. Water a little longer, or when the app suggests it.
Medium confidence
Why: The zone got about 65% of what tomato needed this period. (note 3)

How your farm did this week

Zone	Water given	Water needed	Verdict	Pump runs	Rain
Zone 1 (Tomato)	about 600 L	about 924 L	Too little	1	0 mm
Zone 2 (Tomato)	about 500 L	about 588 L	Watered about right	1	0 mm

Why: The zone got about 65% of what tomato needed this period. (note 3)

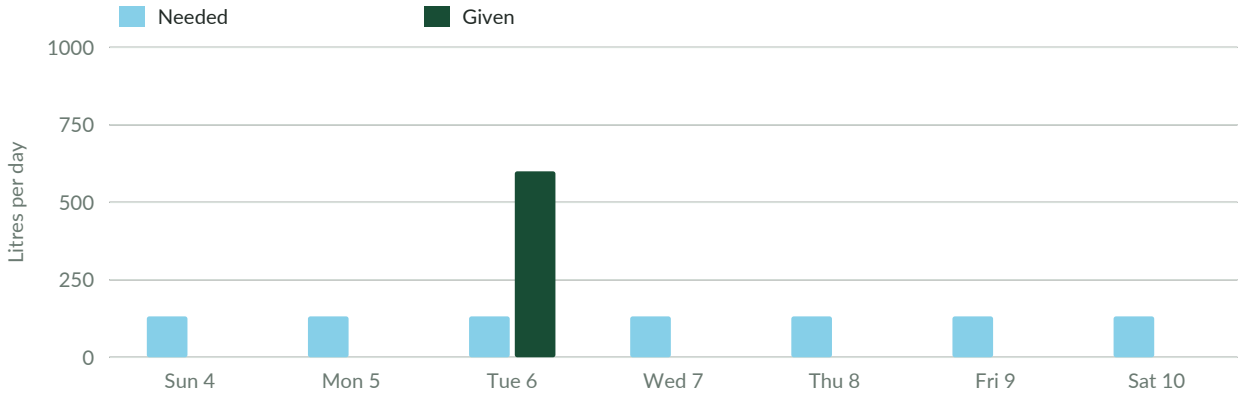
Why: The water given this period was close to what tomato needed. (note 4)

Water Usage Report

Thabo's Farm (demo) · all 2 zones · Sun 4 Oct to Sat 10 Oct
Generated 10 Oct 2026, 07:00 SAST

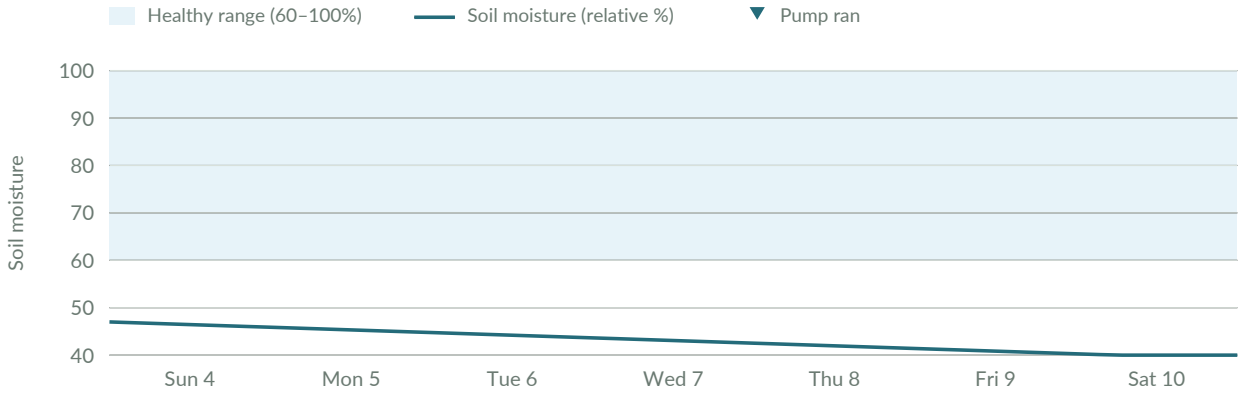
Zone 1 — Tomato

Did my crop get the water it needed?



Not quite - it got about 600 L against about 924 L this week.

Was my soil in the healthy range?



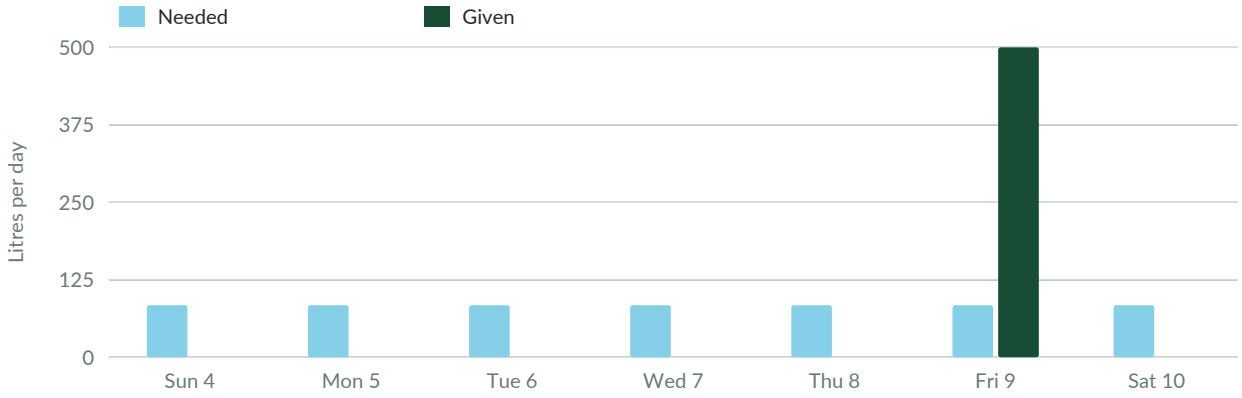
Mostly - it dipped below the healthy range on 7 days this week.

Water Usage Report

Thabo's Farm (demo) · all 2 zones · Sun 4 Oct to Sat 10 Oct
Generated 10 Oct 2026, 07:00 SAST

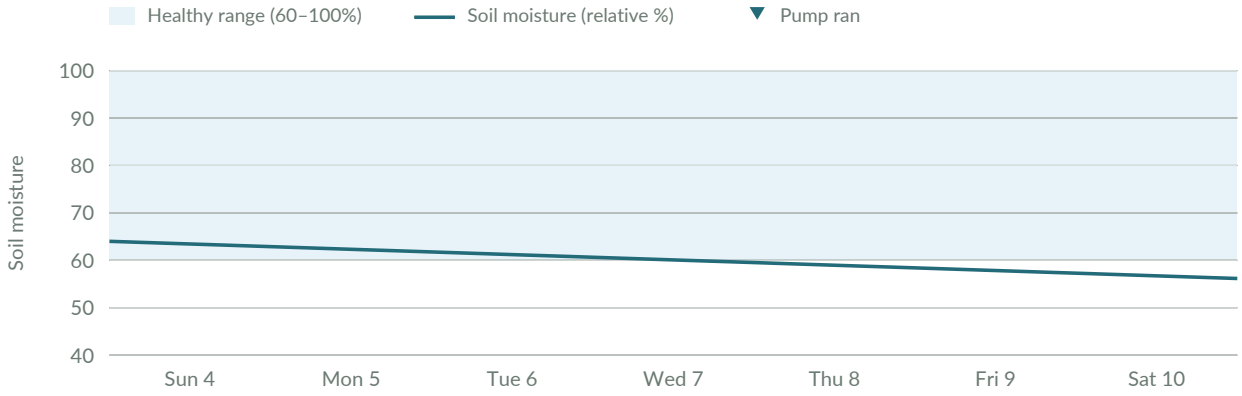
Zone 2 — Tomato

Did my crop get the water it needed?



Yes - about 500 L against about 588 L needed this week.

Was my soil in the healthy range?



Mostly - it dipped below the healthy range on 7 days this week.

What's coming this week?

Sun 11 Oct	Mon 12 Oct	Tue 13 Oct	Wed 14 Oct	Thu 15 Oct	Fri 16 Oct	Sat 17 Oct
31 °C	31 °C	31 °C	31 °C	31 °C	31 °C	31 °C
Rain 0 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm
No watering needed	Water Zone 2	No watering needed	No watering needed	Water Zone 1	No watering needed	Check nearer the day

Area weather vs at your plants

Day	Area high	At the plants	Area humidity	At the plants
Sun 4 Oct	31 °C	31 °C	38%	38%

Litres are estimated (pump running time × flow rate). Soil moisture is relative to this zone's setup.

Water Usage Report

Thabo's Farm (demo) · all 2 zones · Sun 4 Oct to Sat 10 Oct
Generated 10 Oct 2026, 07:00 SAST

Day	Area high	At the plants	Area humidity	At the plants
Mon 5 Oct	31 °C	31 °C	38%	38%
Tue 6 Oct	31 °C	31 °C	38%	38%
Wed 7 Oct	31 °C	31 °C	38%	38%
Thu 8 Oct	31 °C	31 °C	38%	38%
Fri 9 Oct	31 °C	31 °C	38%	38%
Sat 10 Oct	31 °C	31 °C	38%	38%

Only days with readings from a sensor in an irrigation zone are shown.

Water Usage Report

Thabo's Farm (demo) · all 2 zones · Sun 4 Oct to Sat 10 Oct
Generated 10 Oct 2026, 07:00 SAST

Technical appendix

For extension officers and markers. Figures and sources behind the advice; FAO-56 is Allen et al. (1998), page numbers as printed.

Zone 1

Figure	Value	Source
Relative moisture	39%	soil sensor reading 2300 against this zone's legacy ADC wet/dry setup - relative only, not volumetric water content
Soil type	Loam / garden soil	FAO-56 Table 19 (Loam): Table 19 'Loam': theta_FC 0.20-0.30, theta_WP 0.07-0.17. Midpoints used.
Root depth	0.70 m	FAO-56 Table 22, low end of the published range for tomato
Water the soil can hold	91 mm	FAO-56 Eq. 82 from Table 19 soil values and Table 22 root depth
Crop water need factor today	1.05	FAO-56 Eq. 66 across Table 11 stage lengths, Kc values from Table 12 (Kc start: Table 12 group row 'b. Vegetables - Solanum Family')
Reference water use today	6.3 mm	FAO-56 Eq. 6 (Penman-Monteith), inputs from weather_app
This crop's water use today	6.6 mm	FAO-56 Eq. 58: crop factor x reference water use
Dryness it can stand before watering	31 mm	FAO-56 Eq. 83 with the depletion fraction from Table 22, adjusted for today's water use and for soil type
How dry the soil is now	50 mm short of full	FAO-56 Eq. 85 continuous water balance; the point sensor is supporting evidence, not a root-zone volume measurement
TAW	91 mm	FAO-56 Eq. 82, p. 162; Table 19 p. 144; Table 22 pp. 163-165
RAW	30.6 mm	FAO-56 Eq. 83, p. 162
p (adjusted)	0.3	FAO-56 Table 22 p. 163-165; p. 162 ETc adjustment; p. 167 soil adjustment
Kc	1.1	FAO-56 Table 12 pp. 110-114; Eq. 66 p. 132
Confidence	High: your sensor, setup and the weather all agree.	ConfidenceRules (DATA RULE)

Zone 2

Figure	Value	Source
Relative moisture	56%	soil sensor reading 2000 against this zone's legacy ADC wet/dry setup - relative only, not volumetric water content
Soil type	Loam / garden soil	FAO-56 Table 19 (Loam): Table 19 'Loam': theta_FC 0.20-0.30, theta_WP 0.07-0.17. Midpoints used.
Root depth	0.70 m	FAO-56 Table 22, low end of the published range for tomato
Water the soil can hold	91 mm	FAO-56 Eq. 82 from Table 19 soil values and Table 22 root depth
Crop water need factor today	0.67	FAO-56 Eq. 66 across Table 11 stage lengths, Kc values from Table 12 (Kc start: Table 12 group row 'b. Vegetables - Solanum Family')
Reference water use today	6.3 mm	FAO-56 Eq. 6 (Penman-Monteith), inputs from weather_app
This crop's water use today	4.2 mm	FAO-56 Eq. 58: crop factor x reference water use
Dryness it can stand before watering	39 mm	FAO-56 Eq. 83 with the depletion fraction from Table 22, adjusted for today's water use and for soil type
How dry the soil is now	30 mm short of full	FAO-56 Eq. 85 continuous water balance; the point sensor is supporting evidence, not a root-zone volume measurement
TAW	91 mm	FAO-56 Eq. 82, p. 162; Table 19 p. 144; Table 22 pp. 163-165

Water Usage Report

Thabo's Farm (demo) · all 2 zones · Sun 4 Oct to Sat 10 Oct
Generated 10 Oct 2026, 07:00 SAST

Figure	Value	Source
RAW	39.4 mm	FAO-56 Eq. 83, p. 162
p (adjusted)	0.4	FAO-56 Table 22 p. 163-165; p. 162 ETc adjustment; p. 167 soil adjustment
Kc	0.7	FAO-56 Table 12 pp. 110-114; Eq. 66 p. 132
Confidence	High: your sensor, setup and the weather all agree.	ConfidenceRules (DATA RULE)

Applies to every zone

- FAO-56 publishes tomato Kc_end as the range 0.70-0.90 rather than a single value. The midpoint 0.80 is used, so late-season figures carry that spread.

- Stage lengths come from FAO-56 Table 11 (Table 11, Apr/May, Mediterranean), which the book says are indicative only - a real season may run shorter or longer.

- One or more reference-water-use inputs were estimated or defaulted. Technical details retain the input provenance.

- Water amounts in litres are estimates worked out from pump running time and the zone's flow rate. There is no flow meter anywhere in AgroFlow.

Assumptions

- Litres are estimated as pump running time x the zone's flow rate. There is no flow meter on any AgroFlow kit.

- Reference evapotranspiration (ETo) is FAO-56 Penman-Monteith (Eq. 6, p. 24) from the weather app. Solar radiation is always estimated from the day's temperature range (Eq. 50, p. 60); missing humidity is estimated from the temperatures (Eq. 64, p. 124); missing temperatures come from a local field sensor.

- Crop water need is ETc = Kc x ETo (Eq. 58, p. 103), with Kc from Table 12 joined by Eq. 66 (p. 132).

- The daily soil water balance is FAO-56 Eq. 85 (p. 170); stress days are days with Ks < 1 (Eq. 84, p. 169).

- Watering is advised when the root-zone depletion reaches RAW (p. 171): Dr >= RAW.

- Soil sensor readings are relative (wet/dry setup), not a measured volume of water.

- Verdicts: more than 1.2 x the need is too much, less than 0.8 x is too little (VerdictRules, DATA RULE).

Why we said this

1. By the end of today your soil will be drier than tomato can stand, and no rain is expected today.

- Crop: Tomato (this zone's setup)
- Crop stage: growing, day 63 of about 145 (your planting date)
- Soil: Loam / garden soil (this zone's setup)
- How dry the soil is now: about 49.8 mm short of full (your soil sensor and weather app)
- How dry it will be by tonight: about 49.8 mm short of full (weather app and FAO crop water guide)
- Watering point: about 30.6 mm short of full (FAO crop water guide)
- Soil moisture: 39% (your soil sensor)
- Rain expected today: 0 mm (weather app)
- Water your crop will use today: 6.6 mm (weather app and FAO crop water guide)
- How much: about 66.4 min (about 996 L) (this zone's setup)

The rule: We suggest watering once your crop has used about 34% of the water your soil can hold for it, counting any useful rain first.
How sure: High: your sensor, setup and the weather all agree.
What would change it: More rain than forecast today could make this smaller or unnecessary. If it has rained heavily, check the soil before watering.
Technical: Dr 49.8 mm, RAW 30.6 mm, TAW 91 mm, p 0.34, Kc 1.05, ETo 6.3 mm, ETc 6.6 mm. FAO-56 Eq. 58 (p. 103), 82-85 (pp. 162-170). Fao56Engine PrimaryAction WaterNow: projected end-of-day Dr >= RAW (FAO-56 p. 171).

2. At the rate tomato is using water, the soil will reach its watering point in about 2 days.

- Expected watering day: Monday 12 Oct (weather app)
- Crop: Tomato (this zone's setup)
- Crop stage: growing, day 35 of about 145 (your planting date)
- Soil: Loam / garden soil (this zone's setup)
- How dry the soil is now: about 30.3 mm short of full (weather app)

The rule: We look 2 days ahead using the weather forecast, so you can plan.
How sure: High: your sensor, setup and the weather all agree.
What would change it: Rain before Monday would push this back.

Technical: Dr 30.3 mm, RAW 39.4 mm, TAW 91 mm, p 0.43, Kc 0.67, ETo 6.3 mm, ETc 4.2 mm. FAO-56 Eq. 58 (p. 103), 82-85 (pp. 162-170). Forecast projection of Eq. 85 to Dr >= RAW.

3. The zone got about 65% of what tomato needed this period.

- Water given: about 600 L (pump run times)
- Water needed: about 924 L (weather app and FAO crop water guide)
- Verdict: Too little (the rule below)

The rule: More than 1.2 times the need is too much and less than 0.8 times is too little. Pump amounts are estimates from run time and flow rate, so smaller differences are not flagged.
How sure: Medium: water given is estimated from pump run time and the zone's flow rate - there is no flow meter.
What would change it: Longer waterings, or watering when the app suggests it.
Technical: VerdictRules.For(600 L, 924 L) = TooLittle; need = sum of ETc x area (FAO-56 Eq. 58, p. 103).

4. The water given this period was close to what tomato needed.

- Water given: about 500 L (pump run times)
- Water needed: about 588 L (weather app and FAO crop water guide)
- Verdict: About right (the rule below)

The rule: More than 1.2 times the need is too much and less than 0.8 times is too little. Pump amounts are estimates from run time and flow rate, so smaller differences are not flagged.
How sure: Medium: water given is estimated from pump run time and the zone's flow rate - there is no flow meter.
What would change it: A big change in the weather or in how long the pump runs.
Technical: VerdictRules.For(500 L, 588 L) = AboutRight; need = sum of ETc x area (FAO-56 Eq. 58, p. 103).